CLASS 9 — MATHS · FORMULA SHEET

Chapter 11: **World of Algorithms** | Quick revision, one page

Algorithms

- An algorithm is a finite, precise, step-by-step procedure that gives the answer for every valid input and stops.
- Tracing an algorithm means writing the values of the variables after each step in a table.

Column addition

- Add digits column by column from the right: column sum $=$ digit $+$ digit $+$ carry in.
- Digit written $=$ column sum $\bmod 10$; carry out $=$ column sum $\div 10$ (quotient).
- A carry occurs when a column sum is 10 or more; the carry out is at most 1 for two numbers.

Divisors and gcd

- d is a divisor of n if $n \bmod d = 0$. Every $n > 1$ has divisors 1 and n; a prime has exactly 2 divisors.
- gcd(m, n) = the largest number dividing both m and n.
- Euclid's subtraction method: replace the larger of the two numbers by the difference of the two; repeat until both are equal; that number is the gcd.
- Example: $(84, 36) \to (48, 36) \to (12, 36) \to (12, 24) \to (12, 12)$, gcd $= 12$.
- gcd is also the side of the largest square tile that covers an $m \times n$ rectangle exactly.

Remainder (mod)

- $m \bmod n$ is the remainder when m is divided by n: $m = qn + r$ with $0 \le r < n$.
- $m \bmod n = 0$ exactly when n divides m.
- Cyclic patterns (days of the week, clock hours) repeat with period n and are handled by mod n.

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